```
exit 0 <-> [3/5 . exit 1 + 4/5 . back 1]
exit 1 <-> [3/5 . exit 0 + 4/5 . back 0]
back 0 <-> [-4/5 . exit 1 + 3/5 . back 1]
back 1 <-> [-4/5 . exit 0 + 3/5 . back 0]
```

one column for each ground input

input basis patterns

outputs	exit 0	exit 1	back 0	back 1
exit 0	A 0	$\frac{3}{5}$	0	$-\frac{4}{5}$ B
exit 1	$\frac{3}{5}$	0	$-\frac{4}{5}$	0
back 0	0	$\frac{4}{5}$	0	$\frac{3}{5}$
back 1	C $\frac{4}{5}$	0	$\frac{3}{5}$	0 D

Compare inputs `exit 0` and `back 0`

output row	column product
`exit 1`	$(3/5)(-4/5)$
`back 1`	$(4/5)(3/5)$
inner product	0

The other rows contribute zero.

Each squared norm is $9/25 + 16/25 = 1$.